

Assignment-1 (02-01-2023)

- 1) Switch from Notepad++ to vs code IDE.
- 2) VS code overview & Installation
- 3) Basic Practice of HTML source code in VS code.

Assignment-2 (09-01-2023)

- 1) Basic example of css practice in all types of css

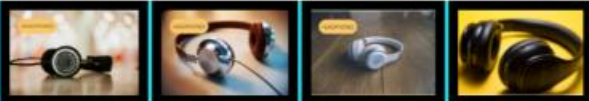


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HTML Business Templates

[Shopping Now](#)





III. HIGH QUALITY

- ORIGINAL DEVICES
- YEAR WARRANTY
- WEEK TO CHECK

II. Visit us and see yourself!

- Soft Skills
- Business Skills
- Presentation

I. About us

- Contact

Find a quality and right headphone

"Success victory"

Life is simple.... Trick to win

Are you want that???

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit.

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Use marquee tag here

Email-Id:

First Name:

Password:

Copyright @ Your student ID

Assignment – 3 (16-01-2023)



OUR FOCUS IS OUR CLIENT SUCCESS!

"Finance Buzz"

Understated Elegance... Unquestionable Taste... Uncompromising Quality...

ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

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Inquiry Form

Email-Id:

First Name:

Password:

Copyright@Your student ID

Assignment-4 (13-02-2023 , 20-02-2023)

Cascading Style Sheet (CSS)

1. Create the following HTML Web page using CSS Internal Style.

This is a Heading

This is a paragraph.

This is another paragraph.

2. Create following HTML Web Page using CSS External file.

THIS IS A HEADING

Lorem Ipsum Dolor Sit Amet, Consectetur Adipiscing Elit, Sed Do Eiusmod Tempor Incidunt Ut Labore Et Dolore Magna Aliqua. Ut Enim Ad Minim Veniam, Quis Nostrud Exercitation Ullamco Laboris Nisi Ut Aliquip Ex Ea Commoda Consequat. Duis Aute Irure Dolor In Reprehenderit In Voluptate Velit Esse Cillum Dolore Eu Fugiat Nulla Pariatur. Excepteur Sint Occaecat Cupidatat Non Proident, Sunt In Culpa Qui Officia Deserunt Mollit Anim Id Est Laborum

3. Create Following Web page to display images using background properties.



4. Create HTML file to design following web page using style.

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Hello World!

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications, is a constituent post-graduate institution of CHARUSAT offering a three-year full-time Master of Computer Applications (MCA) programme with the intake of 180 students per year, a two-year full-time MCA Lateral programme with an intake of 60 students per year, a two-year full-time Master of Science in Information Technology (M.Sc.(IT)) programme with intake of 60 students per year, a five-year full-time Dual Degree Master of Computer Applications (MCA-Dual Degree BCA + MCA) programme with intake of 60 students per year and a Doctor of Philosophy (Ph. D.) programme (Full time/Part time).

The current scenario of industrial and governmental developments in the country provides an ever growing need for highly trained manpower in the field of Computer Applications and Information Technology.

- Coffee
- Tea
 - Black tea
 - Green tea
- Milk

study

the devotion of time and attention to acquiring knowledge on an academic subject, esp. by means of books

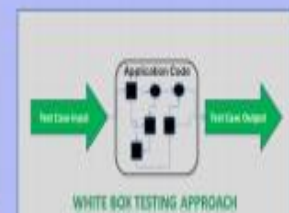
design

a plan or drawing produced to show the look and function or workings of a building, garment, or other object before it is built or made purpose, planning, or intention that exists or is thought to exist behind an action, fact, or material object

business

work

a person's regular occupation, profession, or trade



5. Create following web page using HTML and CSS.

The Bookworm: A Better Book Store

"Old books are best--how tale and rhyme
Float with us down the stream of time!"
-- Clarence Urry, *Old Songs are Best*

The Bookworm Bookshop
1345 Applewood
Springfield, CA 94325
(415) 555-0034

Contents

- [About the Bookworm Bookshop](#)
- [Recent Titles](#)
- [Upcoming Events](#)

About the Bookworm Bookshop

Since 1933, The Bookworm Bookshop has offered rare and hard-to-find titles for the discerning reader. The Bookworm offers:

- Friendly, knowledgeable, and courteous help
- Free coffee and juice for our customers
- A well-lit reading room so you can "try before you buy"
- Four friendly cats: Esmerelda, Catherine, Dulcinea and Beatrice

Our hours are 10am to 9pm weekdays, noon to 7 on weekends.

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Recent Titles (as of July 11, 2012)

- Sandra Bellweather, [Exercise 5 \(Practical List -3\)](#)
- Johnathan Tin, [Exercise 12 \(Practical List -2\)](#)
- Maxwell Burgess, [Image Map](#)
- Alison Caine, [Tables](#)

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Upcoming Events

The Wednesday Evening Book Review
meets, appropriately, on Wednesday evenings at 7 pm for coffee and a round-table discussion. Call the Bookworm for information on joining the group.

The Children's Hour
happens every Saturday at 1 pm and includes reading, games, and other activities. Cookies and milk are served.

Carole Fenney
will be at the Bookworm on Sunday, January 19, to read from her book of poems *Spiders in the Web*.

The Bookworm will be closed
March 1st to remove a family of bats that has nested in the tower. We like the company, but not the mess they leave behind!

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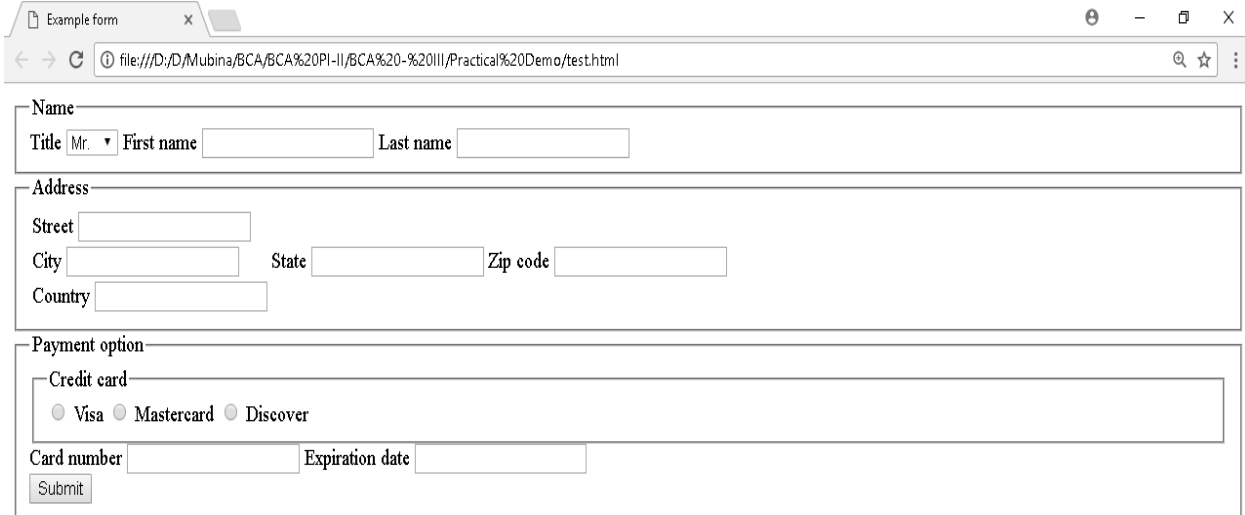
Last Updated: July 11, 2012
Webmaster: Laura Lemay lemay@bookworm.com
© copyright 2012 the Bookworm



Background Color - #CCFFCC

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6. Create a HTML web page to design following form.



The screenshot shows a web browser window with the address bar displaying a file path. The form is titled "Example form" and contains the following sections:

- Name:** A dropdown menu for "Title" with "Mr." selected, followed by text input fields for "First name" and "Last name".
- Address:** Text input fields for "Street", "City", "State", "Zip code", and "Country".
- Payment option:** A section titled "Credit card" containing three radio buttons for "Visa", "Mastercard", and "Discover". Below these are text input fields for "Card number" and "Expiration date", and a "Submit" button.

[Use external style sheet to style above form]



The screenshot shows the same form as before, but now styled with an external CSS sheet. The form is set against a gray background. Each section (Name, Address, Payment option) is enclosed in a rounded rectangular fieldset with a red border. The "Credit card" section is further enclosed in a rounded rectangular legend with a red border. The "Last name" field is highlighted with a red background and yellow text. The "Submit" button is also highlighted with a red background and white text.

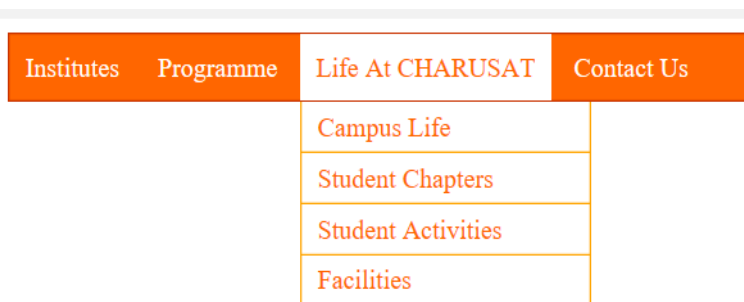
- Body background color - gray
- Fieldset background color - #D3D3D3
- Legend background color - #A9A9A9

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- Mouse over shadow color of an input element- #D35400
- On focus input element background color – red
- On focus Input element text color – yellow;



7. Create Horizontal Menu using HTML & CSS.



Institute: CMPICA, PDPIAS, I2IM, CSPIT, RPCP, MTIN, ARIP

Programme: Undergraduate, Postgraduate, Diploma/Certificate, Research

8. Create Vertical Menu using HTML & CSS.

9. Create following template using HTML and CSS



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Note:

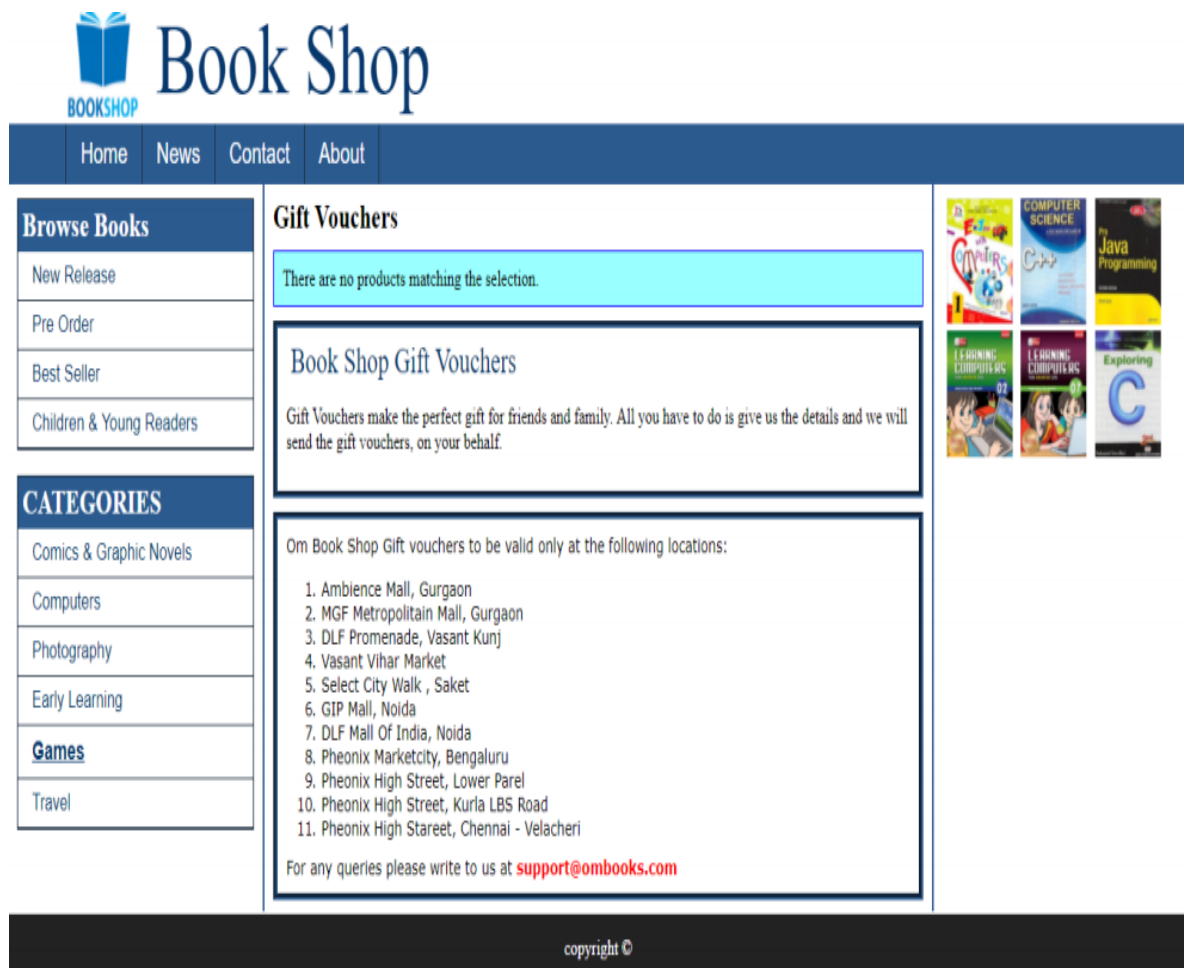
The colours used are as follows:

- Text: #330000
- Body background: #ffffcc
- Header and footer background: #ccaa66
- Left (navigation) column background: #E8D882
- Right (main content) column background: #f1e8b0

The text is displayed in Verdana font.

You should write an external/Internal CSS file. You need to modify the original HTML file only to link to this style sheet, mark up sections (using <div> tags) and set appropriate class and id attributes.

10. Create the following page using HTML & CSS:



The screenshot shows a web page for 'Book Shop'. The header includes a logo and the text 'Book Shop'. Below the header is a navigation bar with links: Home, News, Contact, About. The main content area is divided into three columns. The left column contains a 'Browse Books' section with links: New Release, Pre Order, Best Seller, Children & Young Readers. Below this is a 'CATEGORIES' section with links: Comics & Graphic Novels, Computers, Photography, Early Learning, Games, Travel. The middle column contains a 'Gift Vouchers' section. It starts with a message: 'There are no products matching the selection.' Below this is a section titled 'Book Shop Gift Vouchers' with a description: 'Gift Vouchers make the perfect gift for friends and family. All you have to do is give us the details and we will send the gift vouchers, on your behalf.' Below this is a list of locations where the vouchers are valid: 1. Ambience Mall, Gurgaon, 2. MGF Metropolitan Mall, Gurgaon, 3. DLF Promenade, Vasant Kunj, 4. Vasant Vihar Market, 5. Select City Walk, Saket, 6. GIP Mall, Noida, 7. DLF Mall Of India, Noida, 8. Pheonix Marketcity, Bengaluru, 9. Pheonix High Street, Lower Parel, 10. Pheonix High Street, Kurla LBS Road, 11. Pheonix High Stareet, Chennai - Velacheri. At the bottom of this section is a link: 'For any queries please write to us at support@ombooks.com'. The right column contains a grid of book covers, including 'COMPUTER SCIENCE', 'Java Programming', 'LEARNING COMPUTERS', and 'Exploring C'. The footer contains the text 'copyright ©'.

The colours used are as follows:

Header text :- 06376D

Footer background:- # 222222



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menu background:- 2B5A8E

menu item border:- 213042 menu

font:- Arial, Helvetica, sans-serif active

menu item:- DCE5EE

menu hover:- 2B5A8E

aside border:- 2B5A8E

aside heading:- 2B5A8E

aside list:- 06376D

aside list font:- Arial, Helvetica, sans-serif gift voucher

first article background:- 99FFFF

article border:- 2B5A8E

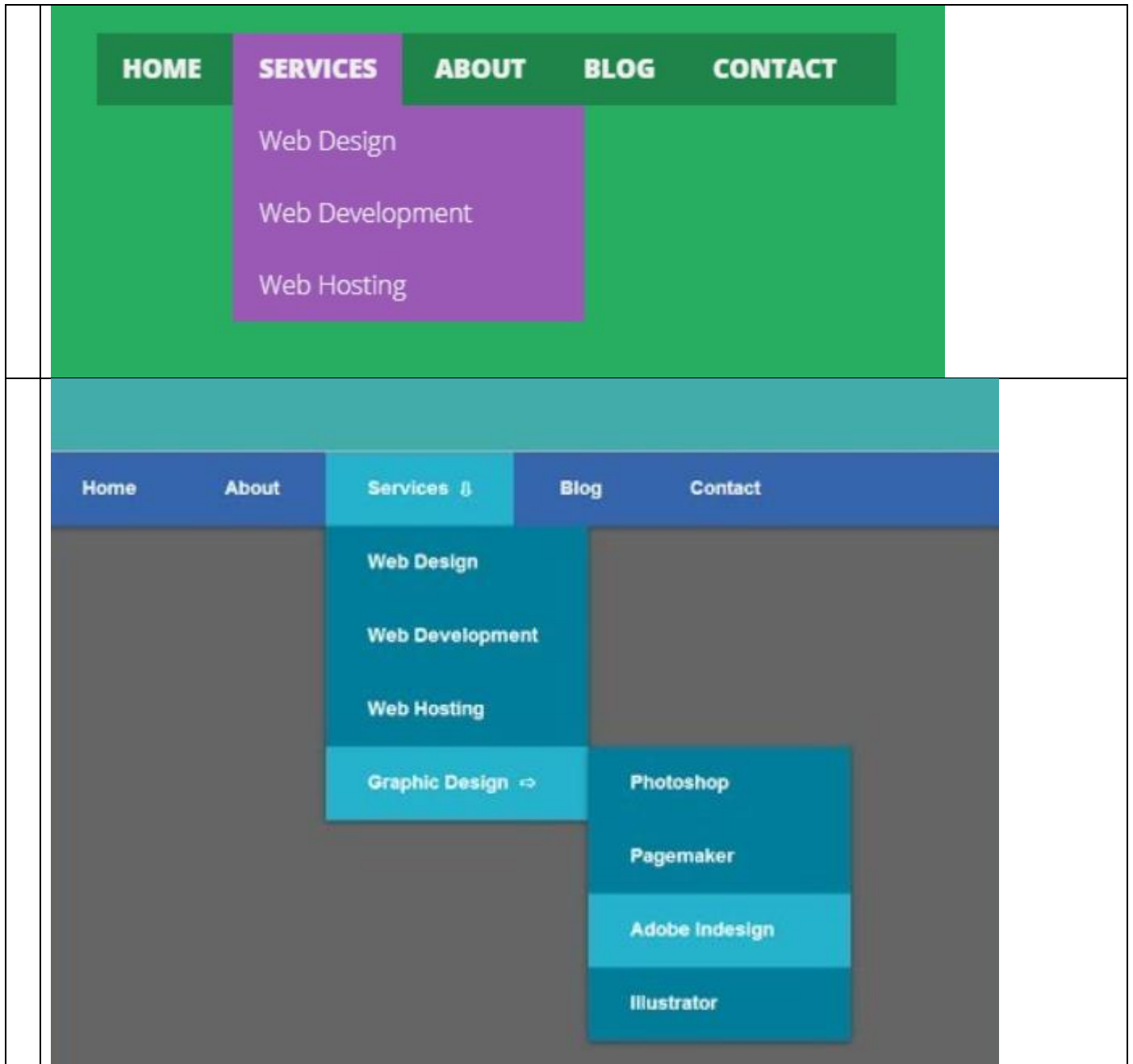
3rd article font:- Verdana, Arial, Helvetica, sans-serif

Note:-

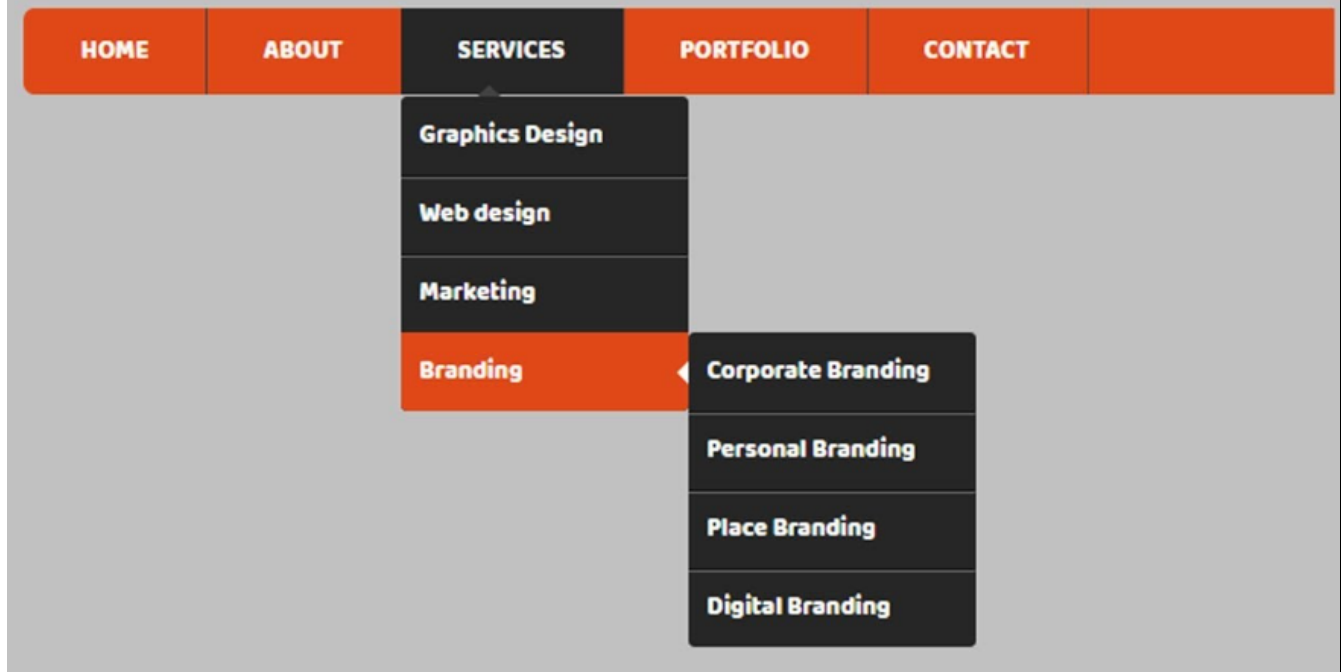
- When user move mouse cursor on aside list item then particular list item display with underline and increase font size
- When user move mouse cursor on image then image should be larger and display border



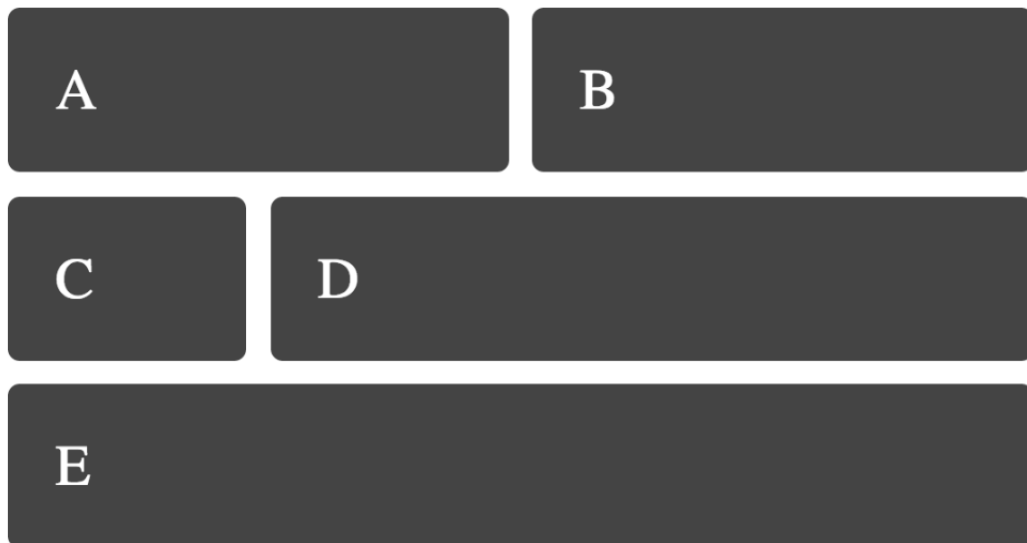
Assignment – 5 (27-02-2023)



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1 Create a webpage that prints below with the use of grid



2 Create a webpage that prints below with the use of grid



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	A	B	
	C	D	

3 Create a webpage that prints below with the use of grid

	A	B	D	
	E	C		
	F			

Lab Assignment || BCA || SEM II || CA130 : Database Fundamentals

Getting familiar with basic SQL Syntax (DDL, DML Statements)

1. Create following table which has following structure:

a. TABLE NAME: EMP

Field Name	Data Type	Null
EMPNO	NUMBER(4) PK	
ENAME	VARCHAR2(20)	
JOB	VARCHAR2(20)	
MGR	NUMBER(4)	YES
HIREDATE	DATE	
SAL	NUMBER(7,2)	
COMM	NUMBER(7,2)	YES
DEPTNO	NUMBER(2) FK	

TABLE NAME : DEPT

Field Name	Data Type	Null
DEPTNO PK	NUMBER(2)	
DNAME	VARCHAR2(20)	
LOC	VARCHAR2(20)	

b. TABLE NAME : SALGRADE

Field Name	Data Type	Null
GRADE	NUMBER(2)	
LOSAL	NUMBER (4)	
HISAL	NUMBER (4)	

2. Insert following records in respective tables.

Table Name: EMP

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	09-DEC-81	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	12-JAN-83	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7802	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	7782	23-JAN-82	1300		10



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Table Name: DEPT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLA
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Table Name: SALGRADE

GRADE	HISAL	LOSAL
1	700	1200
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

3. Solve the following queries.

1.	Find out the name of all employees.
2.	Display all information of employee.
3.	Display all information of employee whose job type is "SALESMAN".
4.	Display all information of employee whose Dept No is 20.
5.	Display employee information whose hire date is in month of DEC.
6.	Display employee information whose salary is greater than 3000.
7.	Find the names of all employees having 'A' as the second letter in their names.
8.	Display department information.
9.	Display Salary Grade Information.
10.	Display Name, Job of employee whose manager name is "BLAKE".
11.	Display Name, Salary of employee whose commission is null.
12.	Display Name, Salary, Hiredate of employee whose Hiredate is in the year of 81.
13.	Update salary by 5% whose department no is 10.
14.	Update commission comm=sal*5% whose job type is "MANAGER".
15.	Display employee name whose name start with 'A'.
16.	Display employee details that have maximum salary.
17.	Display employee details that have minimum salary.
18.	Display average salary department wise.
19.	Display employee details whose salary is between 1000 to 2000.
20.	Display employee information department wise.



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21.	Display your name in upper, lower and title case.
22.	Display all Employee names and their length after trimming them
23.	Extract first 3 characters from your name and display the same.
24.	Extract last 4 characters from your name and display the same.
25.	Extract only 3 to 7 characters of your full name.
26.	Find out the absolute values of the following: -20, 30, 25, -400, -13, -11, 34, -99.
27.	Update the employee table so each employee name will contain XXXXX at the end of the same.
28.	Now display the same such that they won't display XXXXX at the end of each name.
29.	Round the following numbers by 5, 2, 1, 3, 4 decimal places: 4.091978, 232.837733, 822.2325443.

Getting familiar with basic SQL Syntax (DDL, DML Statements)

2. Create following table which has following structure:

a. TABLE NAME: CLIENT_MASTER

Field Name	Data Type	Constraint
CLIENT_NO	VARCHAR2(6)	Primary Key /First letter must Start with 'C'.
NAME	VARCHAR2(20)	Not null
ADDRESS1	VARCHAR2(30)	
ADDRESS2	VARCHAR2 (30)	
CITY	VARCHAR2(15)	
STATE	VARCHAR2 (15)	
PINCODE	NUMBER(6)	
BAL_DUE	NUMBER(10,2)	

a. TABLE NAME : PRODUCT_MASTER

Field Name	Data Type	Constraint
PRODUCT_NO	varchar2(6)	Primary key Must start with P
DESCRIPTION	VARCHAR2(14)	NOT NULL
PROFIT_PERCENT	NUMBER(4,2)	NOT NULL
UNIT_MEASURE	VARCHAR2(10)	NOT NULL
QTY_ON_HAND	NUMBER(8)	NOT NULL
REORDER_LVL	NUMBER(8)	NOT NULL
SELL_PRICE	NUMBER(8,2)	NOT NULL, Cannot be 0
COST_PRICE	NUMBER(8,2)	NOT NULL, Cannot be 0

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a. TABLE NAME : SALESMAN_MASTER

Field Name	Data Type	Null
SALESMAN_NO	VARCHAR2(6)	Primary Key / first letter must start With 'S'
SALESMAN_NAME	VARCHAR2(20)	NOT NULL
ADDRESS1	VARCHAR2(30)	NOT NULL
ADDRESS2	VARCHAR2(30)	
CITY	VARCHAR2(20)	
PINCODE	VARCHAR2(6)	
STATE	VARCHAR2(20)	
SAL_AMT	NUMBER(8,2)	Not Null, cannot be 0
TGT_TO_GET	NUMBER(6,2)	Not Null, cannot be 0
YTD_SALES	NUMBER(6,2)	NOT NULL
REMARKS	VARCHAR2(60)	

Table Name : sales_order

Field Name	Data Type	Null
S_ORDER_NO	VARCHAR2(6)	Primary Key / first letter must start With 'O'
S_ORDER_DATE	DATE	
CLIENT_NO	VARCHAR2(6)	Foreign Key references client_no Of client_master table
DELY_ADDR	VARCHAR2(25)	
SALESMAN_NO	VARCHAR2(6)	Foreign Key references salesman_no Of salesman_master table.
DELY_TYPE	CHAR(1)	delivery : part(P)/full(F), Default 'F'
BILLED_YN	CHAR(1)	
DELY_DATE	DATE	cannot be less than s_order_date
ORDER_STATUS	VARCHAR2(10)	values ('In Process', 'Fulfilled', 'BackOrder', 'Canceled')

Perform Following Changes on Product_Master table

- 1) Change Datatype of Product_no field from number(2) to varchar2(6).
- 2) Assign primary key to product_no field
- 3) Add Constraint to product_no field (Must start with P)

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Table Name : sales_order_details

Field Name	Data Type	Null
S_ORDER_NO	VARCHAR2(6)	Foreign Key references S_order_no of sales_order table.
PRODUCT_NO	VARCHAR2(6)	Foreign Key references Product_no of product_master table.
QTY_ORDERED	NUMBER(8)	
QTY_DISP	NUMBER(8)	
PRODUCT_RATE	NUMBER(10,2)	

Table Name : challan_header

Field Name	Data Type	Null
CHALLAN_NO	VARCHAR2(6)	Primary Key / first two letters must Start with 'CH'.
S_ORDER_NO	VARCHAR2(6)	Foreign Key references s_order_no of Sales_order table.
CHALLAN_DATE	DATE	NOT NULL
BILLED_YN	CHAR(1)	values('Y','N'), Default 'N'.

Table Name : Challan_Details

Field Name	Data Type	Null
CHALLAN_NO	VARCHAR2(6)	Foreign Key references Challan_no of challan_header table.
PRODUCT_NO	VARCHAR2(6)	Foreign Key references Product_no of product_master table.
QTY_DISP	NUMBER(4,2)	NOT NULL

1. . Insert following records in respective tables.

Table Name: CLIENT_MASTER

CLIENT NO.	CLIENT NAME	CITY	PINCODE	STATE	BAL_DUE
C00001	Ivan Bayross	Bombay	400054	Maharashtra	15000
C00002	Vandana Saitwal	Madras	780001	Tamil Nadu	0
C00003	Pramada Jaguste	Bombay	400057	Maharashtra	5000
C00004	Basu Navindgi	Bombay	400056	Maharashtra	0
C00005	Ravi Shreedharan	Delhi	100001		2000
C00006	Rukmini	Bombay	400050	Maharashtra	0



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Table Name: PRODUCT_MASTER

PRODUCT no	DESCRIPTION	PROFIT_PERCENT	UNIT_MEAS U	QTY_ON HAND	REORDER_LVL	SELL_PRICE	COST_PRICE
P00001	1.44 Floppies	5	Piece	100	20	525	500
P03453	Monitors	6	Piece	10	3	12000	11280
P06734	Mouse	5	Piece	20	5	1050	1000
P07865	1.22 Floppies	5	Piece	100	20	525	500
P07868	Keyboards	2	Piece	10	3	3150	3050
P07885	CD Drive	2.5	Piece	10	3	5250	5100
P07965	540 HDD	4	Piece	10	3	8400	8000
P07975	1.44 Drive	5	Piece	10	3	1050	1000
P08865	1.22 Drive	5	Piece	2	3	1050	1000

Table Name: SALES_MASTER

Salesman_No	Salesman_Name	Address1	Address2	City	Pincode	State	Salesamt	Tgt_To_Get	Ytd sales	Remarks
S00001	Kiran	A/14	Worli	Bombay	400002	MAH	3000	100	50	Good
S00002	Manish	65	Nariman	Bombay	400001	MAH	3000	200	100	Good
S00003	Ravi	P-7	Bandra	Bombay	400032	MAH	3000	200	100	Good
S00004	Ashish	A/5	Juhu	Bombay	400044	MAH	3500	200	150	Good

Table Name: SALES_ORDER

S_ORDER_NO	S_ORDER_DATE	CLIENT_NO	DELTYYPE	BILLED	SALESMA NNO	DELYDATE	ORDER_STATUS
O19001	12-JAN-96	C00001	F	N	S00001	20-JAN-96	In Process
O19002	25-JAN-96	C00002	P	N	S00002	27-JAN-96	Canceled
O46865	18-FEB-96	C00003	F	Y	S00003	20-FEB-96	Fulfilled
O19003	03-APR-96	C00001	F	Y	S00001	07-APR-96	Fulfilled
O46866	20-MAY-96	C00004	P	N	S00002	22-MAY-96	Canceled
O10008	24-MAY-96	C00005	F	N	S00004	26-MAY-96	In Process

Table Name: SALES_ORDER_DETAIL

S_ORDER_NO	PRODUCT_NO	QTY_ORDERED	QTY_DISP	PRODUCT_RATE
O19001	P00001	4	4	525
O19001	P07965	2	1	8400
O19001	P07885	2	1	5250
O19002	P00001	10	0	525
O46865	P07868	3	3	3150



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O46865	P07885	3	1	5250
O46865	P00001	10	10	525
O46865	P03453	4	4	1050
O19003	P03453	2	2	1050
O19003	P06734	1	1	12000
O46866	P07965	1	0	8400
O46866	P07975	1	0	1050
O10008	P00001	10	5	525
O10008	P07975	5	3	1050

Table Name: challan_header

CHALLAN_NO	S_ORDER_NO	CHALLAN_DATE	BILLED
CH9001	O19001	12-DEC-95	Y
CH6865	O46865	12-NOV-95	Y
CH3965	O10008	12-OCT-95	Y

Table Name: challan_details

CHALLAN_NO	PRODUCT_NO	QTY_DISP
CH9001	P00001	4
CH9001	P07965	1
CH9001	P07885	1
CH6865	P07868	3
CH6865	P03453	4
CH6865	P00001	10
CH3965	P00001	5
CH3965	P07975	2

QUERIES

Single Table Retrieval :

1.	Find out the names of all the clients.
2.	Print the entire client_master table.
3.	Retrieve the list of names and the cities of all the clients.
4.	List the various products available from the product_master table.
5.	Find the names of all clients having 'a' as the second letter in their names.
6.	Find out the clients who stay in a city whose second letter is 'a'.



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7.	Find the list of all clients who stay in city 'Bombay' or city 'Delhi' or city 'Madras'.
8.	List all the clients who are located in Bombay.
9.	Print the list of clients whose bal_due are greater than value 10000.
10.	Print the information from sales_order table of orders placed in the month of January
11.	Display the order information for client_no 'C00001' and 'C00002'.
12.	Find the products with description as '1.44 Drive and '1.22 Drive.'
13.	Find the products whose selling price is greater than 2000 and less than or equal to 5000.
14.	Find the products whose selling price is more than 1500 and also find the new selling price as original selling price*15.
15.	Rename the new column in the above query as new_price.
16.	Find the products whose cost_price is less than 1500.
17.	List the products in sorted order of their description.
18.	Calculate the square root of the price of each product.
19.	Divide the cost of product '540 HDD' by difference between its price and 100.
20.	List the names, city and state of clients not in the state of 'Maharashtra'.
21.	List the product_no, description, sell_price of products whose description begin with letter 'M'.
22.	List all the orders that were cancelled in the month of March.
23.	Display all the Client names in Capitals and State in Lowercase.
24.	Price must be displayed in 8,2 format. Put '*' before the amount. e.g. ***23.90.
25.	Display Description and Qty_on_hand for each product.
26.	Display sales transactions (qty_ordered and qty_disp) from Sales_order_details.

Set Functions and Concatenation:

27.	Count the total number of orders.
28.	Calculate the average price of all the products.
29.	Calculate the minimum price of products.



Lab Assignment || BCA || SEM II || CA130 : Database Fundamentals

30.	Determine the maximum and minimum product prices. Rename the title as max_price and min_price respectively.
31.	Count the number of products having price greater than or equal to 1500.
32.	Find all the products whose qty_on_hand is less than reorder level.
33.	Print the information of client_master,product_master,sales_order table in the following format for all the records :- {cust_name} has placed order {order_no} on {s_order_date}

Having and Group By:

34.	Print the description and total qty sold for each product.
35.	Find the value of each product sold.
36.	Calculate the average qty sold for each client that has a maximum order value of 15000.00
37.	Find out the total sales amount receivable for the month of jan.It will be the sum total of all the billed orders for the month.
38.	Print the information of product_master,order_detail table in the following format for all the records :- {description} worth Rs. {total sales for the product} was sold.
39.	Print the information of product_master,order_detail table in the following format for all the records :- {description} worth Rs. {total sales for the product} was ordered in the month of {s_order_date in month format}
40.	Find out the total Qty_Dispatch and Qty_ordered of each item from Sales_order_Details.
41.	Find out the no. of sales transactions of each day from Sales_order_details.
42.	Compute average, minimum and maximum Qty_ordered for each product.



Practical List 1 - Bootstrap

Date: (14/12/2022)

Understand the viewport

1. Create sample.html file with one image and paragraph and check the output with and without viewport

Understand the basic bootstrap file structure (Bootstrap 4.6.2)

1. Download Precompiled Library
2. Explore the folders: dist/css, dist/js
3. Explore the folder: Examples and check the output with different screen size using toggle button.

Set Up Bootstrap Library

1. Create Simple bootstrap application with Heading "Welcome to My First Bootstrap Application" and check the source, console
[Note: Import the pre-compiled libraries also check the error in console tab]
2. Create html file to display Headings in container classes. Sample message: "Welcome to My First Bootstrap Page with Container"
[Note: Use Container and Container-fluid class]
3. Check the output of above web page in different devices and orientation.
[Note: Use the toggle button]

Date: (21/12/2022)

1. Explore various bootstrap typography classes.
2. Explore various text and background classes with the heading “Smt. Chandaben Mohanbhai Patel Institute of Computer Application – CMPICA, CHARUSAT, Changa”
3. Create html page to create your profile with using bootstrap typography classes, bootstrap text and background colors. [Apply combination of text and background colors]
4. Create a bootstrap page to create three equal column grid.
[Note: apply both container and fluid container class for different view]



Laboratories

All laboratories are well equipped with state of art equipments, which not only cater to UG/PG programmes, but are also very useful for testing and consultancy projects. The details are available in respective departments. The laboratories provide students with opportunities to verify the theoretical knowledge that they imbibe in the classroom, and experience scientific principles first-hand.



Library for Students

The Knowledge Resource center (Central Library) – a proud partner in the institute's march towards its vision, plays a vital role in acquisition, organization and dissemination of knowledge. It has put in place policies and procedure systems and services and the ambience that facilitate creation of new knowledge.

It has an excellent collection of both print and electronic books, journals, technical reports, back volumes and other reading material. It has adequate infrastructure to meet its requirements, has computerized all its operation using software developed in-house, and provide access to the collection through Online Public Access Catalogue (OPAC).



Sports

CHARUSAT's cricket and football teams have won several state level championships. The university campus has facilities for indoor and outdoor games like badminton, table tennis, cricket, volleyball and football.

Date: 28/12/2022

5. Create a bootstrap page to create following layout.

My First Bootstrap Page

Resize this responsive page to see the effect!

CHARUSAT

Vision

To become a dynamic global institution in a knowledge driven world through excellence in teaching, research and social contributions

Mission

To serve society by striving to transform it through creation, augmentation, dissemination and perpetuation of knowledge

CMPICA

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications, is a constituent post-graduate institution of CHARUSAT offering a three-year full-time BCA programme, B.Sc.(IT) programme, a two-year full-time MCA Lateral programme, M.Sc. (IT) programme & a (Ph.D.) programme (Full time/Part time).

BCA-4

- CA220- Basics of Web Designing
- CA221- Data Structures & Algorithms
- CA222- Advanced Database System
- CA223- Fundamentals of Computer Networks
- CA224- Creativity, Problem Solving and Innovation

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

6. Create a bootstrap page to create three unequal column grid /CSS Media Query Layout.

Welcome To CMPICA

BCA - 4 - CA220- Basics of Web Designing - CA221 Data Structures & Algorithms - CA222 Advanced Database System - CA223 Fundamentals of Computer Networks - HS133 C Creativity, Problem Solving and Innovation	CA220: Basics of Web Designing (200 Marks) Contact Hours: 07 Objective: This syllabus is designed to familiarize students with Responsive web designing, HTML, CSS and JavaScript framework – AngularJS. Pre-requisite: CA113 - Programming the Internet-I and CA213 - Programming the Internet-II. Methodology & Pedagogy: During theory lectures students will be introduced to the fundamental basics of Responsive web design, Media Query and AngularJS framework. During Practical sessions students will be required to design and develop responsive web sites using HTML5, CSS3 and AngularJS. Learning Outcomes: Upon successful completion of the course, the students will be able to deliver an optimal user experience to all devices including tablets, smartphones, feature phones, laptops, and large screens.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="background-color: #FFD700; color: black;">Faculty :- Mubina Malik & Swati Patel</td> <td style="background-color: #0000FF; color: white;">Course Credit :- 7 (4 + 3)</td> </tr> </table> Internal Evaluation - Unit test :- 08 Marks - Sessional Exam :- 07 Marks - Attendance :- 05 Marks - Case Study (Group of 3 Students) :- 10 Marks	Faculty :- Mubina Malik & Swati Patel	Course Credit :- 7 (4 + 3)
Faculty :- Mubina Malik & Swati Patel	Course Credit :- 7 (4 + 3)			

© Developed By 17BCAXXX ©

Body :- background-color: bisque

Date: 4/01/2023

7. Create a bootstrap page to design following layout for 3 different screen size. (for Large Screen, Small Screen, Extra small screen)



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BCA-4

- CA220- Basics of Web Designing

For Extra small screen



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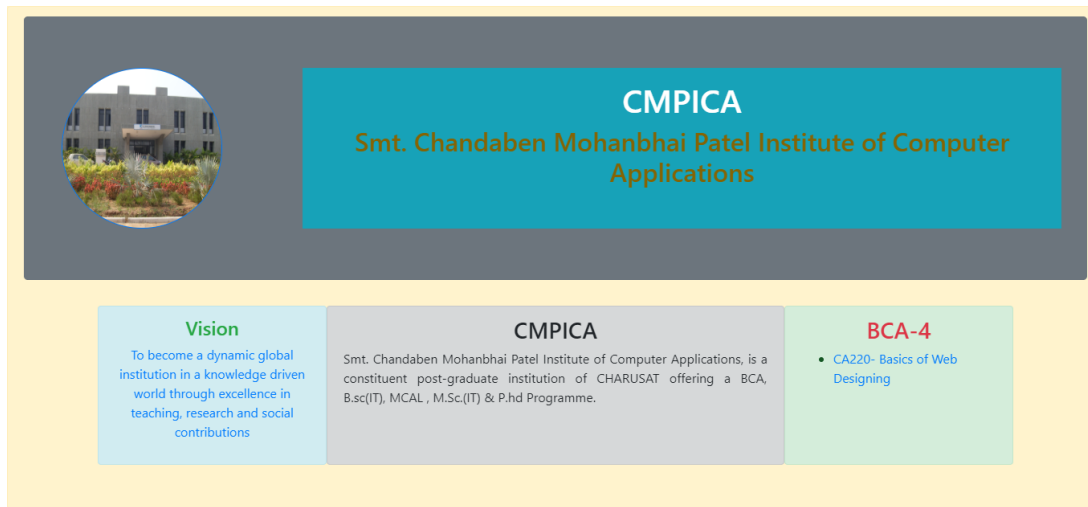
Smt. Chandaben Mohanbhai Patel Institute of Computer Applications, is a constituent post-graduate institution of CHARUSAT offering a BCA, B.sc(IT), MCAL, M.Sc.(IT) & P.hd Programme.

BCA-4

- CA220- Basics of Web Designing

For small screen

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing



For
Large

screen

8. Create a bootstrap page to design following layout.

Welcome To FOOD ZONE

Food Zone Chinese Restaurant, located in the Village of Dexter, MI just west of Ann Arbor, serves a mixture of authentic and traditional Chinese dishes. Please surf around our site for menus, specials, location, hours, and educational information. We are located a couple of miles east of downtown Dexter in the Country Market Shopping Center at:

Food Zone Chinese Restaurant, located in the Village of Dexter, MI just west of Ann Arbor, serves a mixture of authentic and traditional Chinese dishes. Please surf around our site for menus, specials, location, hours, and educational information. We are located a couple of miles east of downtown Dexter in the Country Market Shopping Center at:

Address: 7023 Dexter-Ann Arbor Rd. Dexter, MI 48130 (734) 426-3410

Operating Hours

Monday - Thursday:	10:30 AM to 9:30 PM
Friday - Saturday:	10:30 AM to 10:30 PM
Sunday:	11:00 AM to 9:30 PM



- Services
- Find My Food
- Blog
- Offers
- Contact

Biggest Offer of the day!!!!

[Click to view offers](#)

Know more about the special events

[Click to know special events](#)



Yummy treat for my family..

Our kids had the best time at this place. They could eat all the chatpata items.. they just loved it. Also the view is amazing.. wonderful.. Kids loved the water sports activities being performed on the river.. [Read More](#)



Best garden restaurant with view..

The food zone is nicely developed on the bank of Tapi River. The choice of vast menu is great. We loved the food and the view. The breeze of air coming from the river felt amazing. It has the system of self-service. We need to buy the coupons and than take the food... [Read More](#)



Hot Drinks	
Cafe Latte	100
Cafe Mocha	250
Cafe Americano	150
Sandwiches	
Crispy Veg Wrap	100
Tandoori Paneer	250
Spinach 'N' Corn Cheese	300

© FoodZone

For large Screen (lg)

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

Welcome To FOOD ZONE

Food Zone Chinese Restaurant, located in the Village of Dexter, MI just west of Ann Arbor, serves a mixture of authentic and traditional Chinese dishes. Please surf around our site for menus, specials, location, hours, and educational information. We are located a couple of miles east of downtown Dexter in the Country Market Shopping Center at:

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Hot Drinks

Cafe Latte	100
Cafe Mocha	250
Cafe Americano	150

Sandwiches

Crispy Veg Wrap	100
Tandoori Paneer	250
Spinach 'N' Corn Cheese	300

FOODZONE

Services

Find My Food

Blog

Offers

Contact

Biggest Offer of the day!!!!

[Click to View offers](#)

X

Know more about the special events

[Click to know special events](#)

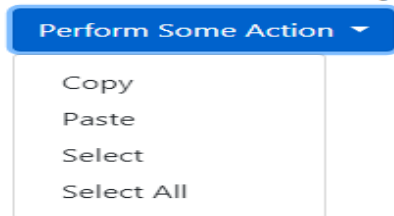
X

© FoodZone

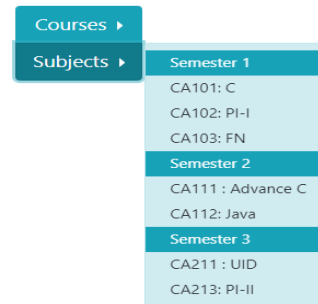
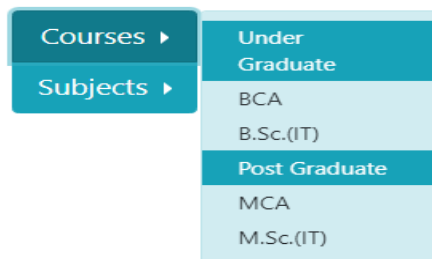
For small Screen (sm)

Date: 11/1/2023

9. Design below Bootstrap Dropdown Menu as given.



Grid Example Exercises



Date: 18/1/2023

Design below page to display collapsible content using bootstrap Card

10. Create Collapsible card for Menu (Note: Use Card to display content)

JQuery
HTML
CSS

Introduction: JQuery

JQuery is a lightweight, "write less, do more", JavaScript library. The purpose of JQuery is to make it much easier to use JavaScript on your website. JQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into methods that you can call with a single line of code. JQuery also simplifies a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

last updated on 20th December,2021

Introduction: HTML

HTML stands for Hyper Text Markup Language HTML is the standard markup language for creating Web pages HTML describes the structure of a Web page HTML consists of a series of elements HTML elements tell the browser how to display the content HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

last updated on 20th December,2021

Introduction: CSS

CSS stands for Cascading Style Sheets CSS describes how HTML elements are to be displayed on screen, paper, or in other media CSS saves a lot of work. It can control the layout of multiple web pages all at once External stylesheets are stored in CSS files

last updated on 20th December,2021

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

11. Create **Collapsible tabs** for given Menu (Note: Use Card to display content)

JQueryHTMLCSS

Introduction: JQuery

JQuery is a lightweight, "write less, do more", JavaScript library. The purpose of JQuery is to make it much easier to use JavaScript on your website. JQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into methods that you can call with a single line of code. JQuery also simplifies a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

last updated on 20th December,2021

Bootstrap Nav Example:

12. Create a bootstrap page with horizontal navigation.

CHARUSAT

About CHARUSATCSPITCMPICADEPSTARRPCPPDPISIZIM

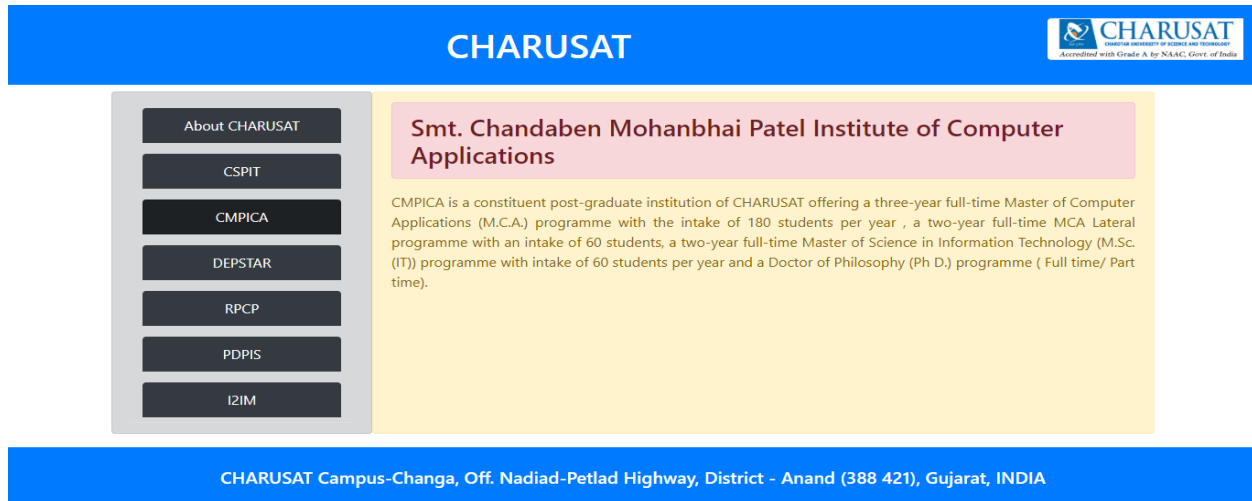
Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

CMPICA is a constituent post-graduate institution of CHARUSAT offering a three-year full-time Master of Computer Applications (M.C.A.) programme with the intake of 180 students per year , a two-year full-time MCA Lateral programme with an intake of 60 students, a two-year full-time Master of Science in Information Technology (M.Sc.(IT)) programme with intake of 60 students per year and a Doctor of Philosophy (Ph D.) programme (Full time/ Part time).

CHARUSAT Campus-Changa, Off. Nadiad-Petlad Highway, District - Anand (388 421), Gujarat, INDIA

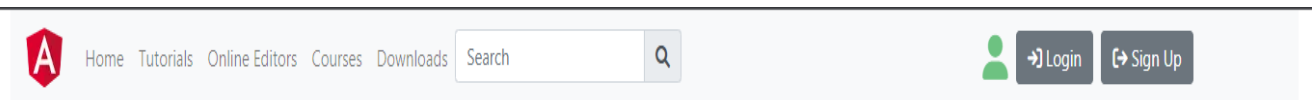


13. Create a bootstrap page with vertical navigation.



Navigation Bar Example:

14. Create navigation bar with toggler button for medium size screen device



Bootstrap Form Example:

15. Create Full width form as given below

Forms in Bootstrap

Username:

Password:

16. Create inline form as given below

Bootstrap inline form login form

Email: Password: ☐ Remember me



Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

17. Create Credit Card Payment form as given below (Note: Use Form Grid to design the form)

Credit Card Payment

CVC code is required.

Card Holder's Name

Card Number

Card Exp. Date

01

2018

CVC

Amount

\$

39

.00

Cancel

Submit

18. Create form as given below (Note: Use Form Grid to design the form & Apply validation classes)

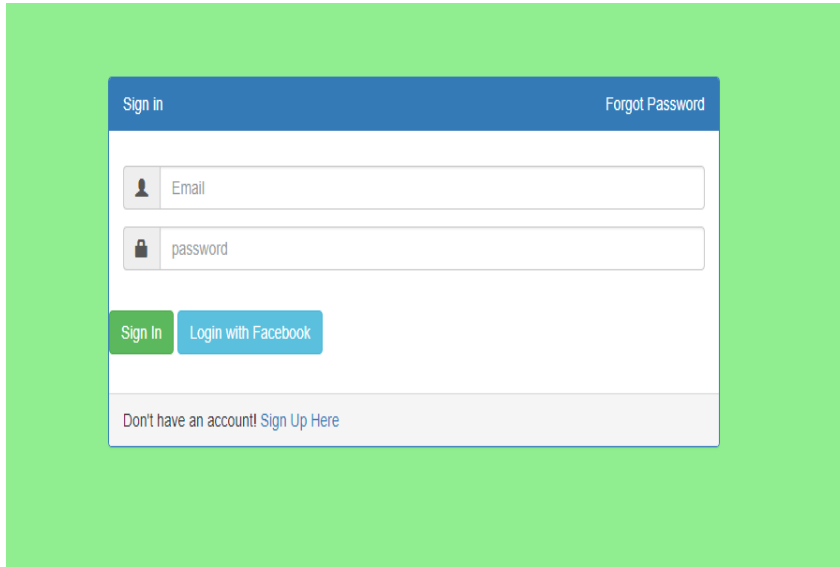
Complex Form Example

Account #	Control #	Amount
XXXXXXXXXXXXXXXXXX	0000	\$ Amount
First Name	Last Name	
City	State	Postal Code
	Pick a state	
Color	Message	
☒ Red ☐ Green ☐ Blue		
Save Changes Cancel		

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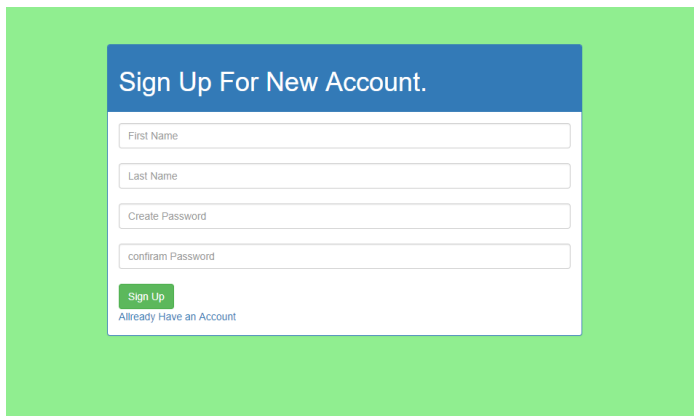
19. Create following web page to display Navigation Bar with login & Sign up options.

19.1 -> Login Page with validation



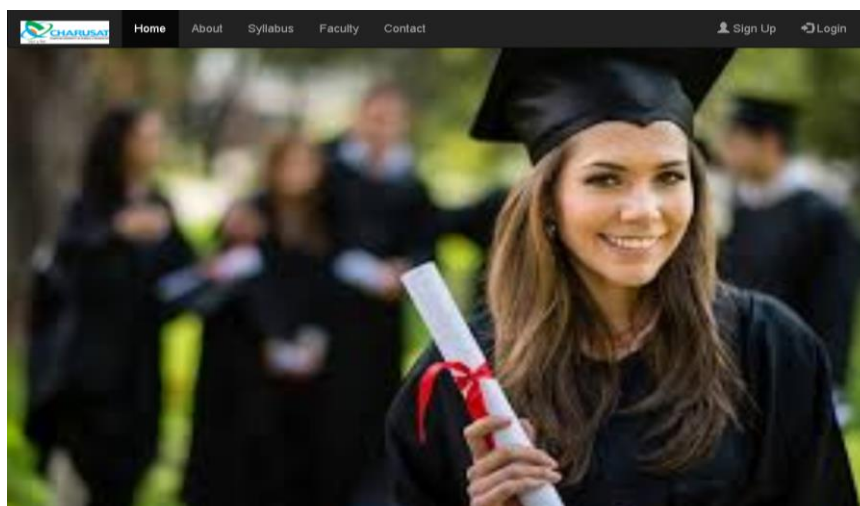
A login page design with a blue header bar containing "Sign in" and "Forgot Password" links. Below the header are two input fields: "Email" with a person icon and "password" with a lock icon. There are two buttons: "Sign In" (green) and "Login with Facebook" (blue). At the bottom, there is a link: "Don't have an account! Sign Up Here".

19.2 -> Sign up (with validation)



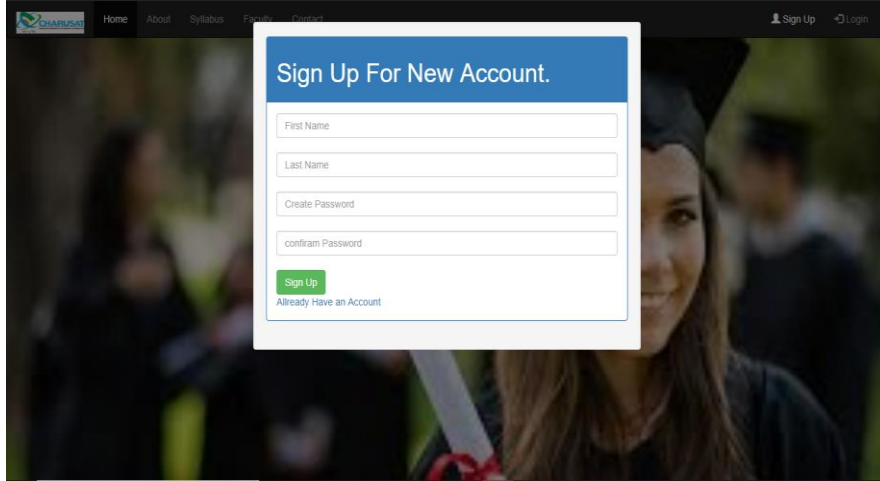
A sign up page design with a blue header bar containing "Sign Up For New Account." Below the header are four input fields: "First Name", "Last Name", "Create Password", and "confirm Password". There are two buttons: "Sign Up" (green) and "Already Have an Account" (blue).

19.3 -> Navigation Bar



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19.4 -> Modal to link Sign Up and Login form



Sign Up For New Account.

First Name

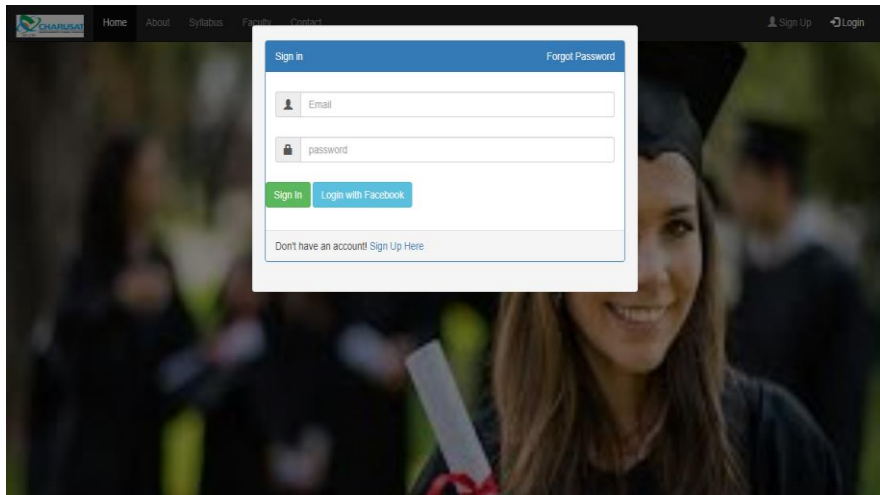
Last Name

Create Password

confirm Password

[Sign Up](#)

[Already Have an Account](#)



Sign in [Forgot Password](#)

[Sign in](#) [Login with Facebook](#)

[Don't have an account? Sign Up Here](#)



Practical List 2 - AngularJS

1.	Create an AngularJS application to display User Name inputted value in heading with welcome message.
2.	Create an AngularJS application which take two numbers as an input and perform arithmetic operations.
3.	Create an AngularJS application Which display message, change background color and text color of message. Enter Message:- Hello Enter Background Color: green Enter Text Color: white You wrote:- Hello
4.	Create an AngularJS application to display list of subjects. Also take input as position of subject and display respective subject name. Note: use array to store subjects list. Enter Poistion:- 1 1 subject is = BWD List of Subjects 1. DSA 2. BWD 3. ADBMS 4. ACN
5.	Create an AngularJS application Which display students details according to student ID inputted in textbox. Note - [Take data as ID, Name, Course, Semester, City, Sub1Marks, Sub2Marks, Sub3Marks] Note: Use Array with object type.
6.	Create an AngularJS application which will display Student ID and Full name, Total marks of 5 subjects. Note: Take object type variable for StudentID, FirstName, LastName and ObtMarks (For 5 subjects)
7.	Create an AngularJS application to display "Welcome CMPICA" message in heading Using Module and controller.
8.	Create an AngularJS application to display user inputted value as "Welcome {{username}}" in heading Using Module and controller. [Take Submit Button to display message]
9.	Create an AngularJS application to calculate Addition, Subtraction, Division and Multiplication of two number using Click events and also change textbox background color when user focus and lost textbox area.[Note: take two number and four button]
10.	Create an AngularJS application which display Heading background color when user enter and leave the heading area.
11.	Create an AngularJS application to increment the count variable when user press any key of keyboard and change text color of count variable when user release the key.
12.	Create an AngularJS application to search Product based on product id form the list of product, also display the details of Product such as (Product ID, Product Name, Product Price, Product Brand and Product Image)

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

	Search Product Enter Product ID:- P001 Product Infomation Product Id P001 Product Name Mobile Product Price 25000 Product Image 														
13.	Use the above example for displaying product detail according to user selected product name from the dropdown menu. If user select product name respective details will be display in proper format. [Use bootstrap to design proper format] Search Product Enter Product Name:- Mobile <table><thead><tr><th>ProductId</th><th>ProductName</th><th>Price</th></tr></thead><tbody><tr><td>P001</td><td>Mobile</td><td>25000</td></tr></tbody></table> Mobile Mobile TV AC Microwave WashingMachine Laptop 	ProductId	ProductName	Price	P001	Mobile	25000								
ProductId	ProductName	Price													
P001	Mobile	25000													
14.	Create an AngularJS application to Calculate Result of selected student’s ID from dropdown menu. Also display the student ID according to city in a dropdown menu. [Assume three subject marks] Student Result Select Student ID:- 17BCA095 <table><thead><tr><th colspan="2">Name:- 17BCA095 Hiral</th></tr></thead><tbody><tr><td>PI-II</td><td>55</td></tr><tr><td>OOCp</td><td>62</td></tr><tr><td>FCN</td><td>80</td></tr><tr><td>Total</td><td>197</td></tr><tr><td>Percentage</td><td>66</td></tr><tr><td>Class</td><td>second Class</td></tr></tbody></table>	Name:- 17BCA095 Hiral		PI-II	55	OOCp	62	FCN	80	Total	197	Percentage	66	Class	second Class
Name:- 17BCA095 Hiral															
PI-II	55														
OOCp	62														
FCN	80														
Total	197														
Percentage	66														
Class	second Class														
15.	Create an AngularJS application to display above student details according to sorting order selected by user from radio button. [Note: take four radio button for sorting student ID and student Name as ascending and descending order]														

Lab Assignment || BCA || SEM IV || CA220: Basics of Web Designing

16. Create Login Application which allow user to input login ID and password. Perform the below operation according to authentication.

1. Check user have clicked Login or Logout button
2. On Click of Logout button display "Thank you for Visiting site" message
3. On click of Login button display content according to LoginID.
 - If Login ID is student id then allow student to view his/her information with result
 - If login ID is Admin, then allow Admin to view entire student details in proper format, Also allow Admin to Add and Delete the student details

On Page Load

Login

If user clicked Login with Login ID as student display information

Name:- 17BCA095 Hiral	
PI-II	55
OOCP	62
FCN	80
Total	197
Percentage	66
Class	second Class

If user clicked Login with Login ID as Admin then display information

Login

Welcome Admin

Student ID	Name	City	Course	BWD Marks	DS Marks	SAD Marks	Total	Percentage	Class	
20BCA003	PINAL	surat	BCA	70	67	87	224	74.67	Dist	Delete
20MCA002	POOJA	baroda	MCA	60	67	77	204	68.00	First	Delete
20MCA001	URMISH	anand	MCA	60	47	57	164	54.67	Pass	Delete
20BSC020	HARDIK	anand	BSC(IT)	40	37	47	124	41.33	Pass	Delete
19MCA003	KAMAL	anand	MCA	80	56	77	213	71.00	Dist	Delete
19BCA001	SHREYA	ahmedabd	BCA	90	97	97	284	94.67	Dist	Delete
										Add

17. Create an AngularJS application to display data in sorting order by clicking on table header and also filter data based on price.



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Click the table headers to change the sorting order:

Enter Price to Filter:

Product Id	Product Name	Price	date
P003	AC	Rs 30,000.00	December 3, 2016
P001	MOBILE	Rs 25,000.00	October 23, 2014
P005	WASHINGMACHINE	Rs 45,000.00	February 15, 2011
P006	LAPTOP	Rs 35,000.00	December 13, 2015
P002	TV	Rs 55,000.00	July 13, 2010
P004	MICROWAVE	Rs 15,000.00	August 3, 2012

Click the table headers to change the sorting order:

Enter Price to Filter:

Product Id	Product Name	Price	date
P003	AC	Rs 30,000.00	December 3, 2016
P006	LAPTOP	Rs 35,000.00	December 13, 2015

ASSIGNMENT-1

Objectives

- Hands on with Basics of Data Definition Language (DDL) commands
- Hands on with Basics of Data Manipulation Language (DML).
- To understand and apply constraints to the tables.
- To apply JOINS and GROUPBY with aggregate functions.

a. TABLE NAME: EMP

Field Name	Data Type	Null
EMPNO	NUMBER(4)	
ENAME	VARCHAR2(10)	
JOB	VARCHAR2(9)	
MGR	NUMBER(4)	
HIREDATE	DATE	
SAL	NUMBER(7,2)	
COMM	NUMBER(7,2)	
DEPTNO	NUMBER(2)	

TABLE NAME : DEPT

Field Name	Data Type	Null
DEPTNO	NUMBER(2)	
DNAME	VARCHAR2(14)	
LOC	VARCHAR2(13)	

b. TABLE NAME : SALGRADE

Field Name	Data Type	Null
GRADE	NUMBER(2)	
LOSAL	NUMBER (4)	
HISAL	NUMBER (4)	



1. Insert following records in respective tables.

Table Name: EMP

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	09-DEC-81	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	12-JAN-83	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7802	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	7782	23-JAN-82	1300		10

Table Name: DEPT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLA
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Table Name: SALGRADE

GRADE	HISAL	LOSAL
1	700	1200
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

2. Solve the following queries.

1.	Find out the name of all employees.
2.	Display all information of employee.
3.	Display all information of employee whose job type is "SALESMAN".
4.	Display all information of employee whose Dept No is 20.
5.	Display employee information whose hiredate is in month of DEC.



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6.	Display employee information whose salary is greater than 3000.
7.	Find the names of all employees having 'A' as the second letter in their names.
8.	Display department information.
9.	Display Salary Grade Information.
10.	Display Name, Job of employee whose manager name is "BLAKE".
11.	Display Name, Salary of employee whose commission is null.
12.	Display Name, Salary, Hiredate of employee whose Hiredate is in the year of 81.
13.	Update salary by 5% whose department no is 10.
14.	Update commission comm=sal*5% whose job type is "MANAGER".
15.	Display employee name whose name start with 'A'.
16.	Display employee details that have maximum salary.
17.	Display employee details that have minimum salary.
18.	Display average salary department wise.
19.	Display employee details whose salary is between 1000 to 2000.
20.	Display employee information department wise.
21.	Display your name in upper, lower and title case.
22.	Display all Employee names and their length after trimming them
23.	Extract first 3 characters from your name and display the same.
24.	Extract last 4 characters from your name and display the same.
25.	Extract only 3 to 7 characters of your full name.
26.	Find out the absolute values of the following: -20, 30, 25, -400, -13, -11, 34, -99.
27.	Update the employee table so each employee name will contain XXXXX at the end of the same.
28.	Now display the same such that they won't display XXXXX at the end of each name.
29.	Round the following numbers by 5, 2, 1, 3, 4 decimal places: 4.091978, 232.837733, 822.2325443.



ASSIGNMENT-2

Objective:

- To describe constant, variable, variable initialization, and basic operations on the variables in PL/SQL.

1. PL/SQL Block

- 1 Write a PL/SQL block to print 'WELCOME' message.
- 2 Write a PL/SQL block to input and to print your personal detail in the following format using appropriate variable.
Name: <<Name>>
Address: <<Address>>
City: <<City>>
Mobile: <<Mobile Number>>
Birth Date: <<Birth date>>
- 3 Write a PL/SQL block to print to initialize two number variables and print the total of it.
- 4 Write a PL/SQL block to swap the values of two number type of variables. Print the values of variables before and after the swap.
- 5 Write a PL/SQL block to print minimum of two numbers.
- 6 WRITE A PL/SQL BLOCK TO FIND MAX OF THREE NUMBERS.
- 7 WRITE A PL/SQL BLOCK TO PRINT GRADE OF THE STUDENT
INPUT : ROLLNO, MARKS1, MARKS2, MARKS3
CALCULATE TOTAL = MARKS1 + MARKS2 + MARKS3 / 3
BASED ON TOTAL CALCULATE PER
GRADE IS CALCULATED AS PER FORMULA
IF PER >80 GRADE IS AA
IF PER BETWEEN 70 AND 80 GRADE IS A
IF PER BETWEEN 60 TO 70 GRADE IS AB
IF PER BETWEEN 50 TO 60 GRADE IS B
IF PER BETWEEN 40 TO 50 GRADE IS C
IF PER LESS THAN 40 GRADE IS FAIL
- 8 WRITE A PL/SQL BLOCK TO PRINT 10 NUMBERS USING WHILE AND FOR LOOP
- 9 WRITE A PL/SQL BLOCK TO PRINT MULTIPLICATION TABLE
USING WHILE AND FOR LOOP
- 10 WRITE A PL/SQL BLOCK TO PRINT FACTORIAL OF GIVEN NUMBER.



ASSIGNMENT-3

Objective:

- To describe discuss DML statement in PL/SQL with exception handling also.

PL/SQL Implicit Cursor

- 1 Write a PL/SQL block to display name and mobile number from EMP table whose id is 101.
- 2 Write a PL/SQL block to insert 5 numbers in temp table, update the any number by taking input from user, delete any number as per user choice. Also print appropriate message.
- 3 WRITE A PL/SQL BLOCK TO INSERT ODD NUMBERS IN T1 TABLE AND EVEN NUMBERS IN T2 TABLE
WRITE A PL/SQL BLOCK TO DELETE VALUES FROM TABLE T1
WRITE A PL/SQL BLOCK TO DELETE LAST 50 RECORD OUT OF 100 FROM TEMP TABLE
- 4 Create table of patient with fields
PATIENT(pid integer, pnm varchar2(10), age integer)
---> TAKE PATIENT ID AS PRIMARY KEY,
AGE SHOULD BE GREATER THAN ZERO
ADD GENDER FIELD INTO PATIENT TABLE AND UPDATE THE RECORD

Write a PL/SQL block which will print
the patient report in the following format
INPUT PATIENT ID FROM USER AND
COMPARE IT WITH TABLE ID AND DISPLAY ACCORDINGLY

patient id: XXX Patient name: XXX
AGE <20 : PRINT IT
AGE >= 20 : PRINT IT
AGE >20 and AGE <=30 : PRINT IT
AGE > 30 & AGE <=40 : PRINT IT
AGE >40 & AGE <60 : PRINT IT
AGE >=60 : DISPLAY IT

- 5 WRITE A PL/SQL BLOCK TO DO THE FOLLOWING:
INPUT 20 NUMBERS FROM USER
A) FIRST ADD ALL NUMBERS IN T3 TABLE
B) IF NUMBER IS 5,10,15 THEN DO UPDATE T3 (ADD 20 IN)
ELSE DELETE OTHER RECORDS
Print appropriate message for valid and invalid record.



ASSIGNMENT-4

Objective:

- To describe SELECT-into clause – only one output statement in PL/SQL

```
CREATE TABLE DOCTOR(DID INTEGER PRIMARY KEY, DNAME  
VARCHAR2(10))
```

```
CREATE TABLE PATIENT(pid integer PRIMARY KEY,  
patientnm varchar2(10),  
age integer CHECK(AGE > 0),  
GENDER CHAR(1) CHECK(GENDER IN ('M','F','m','f')),  
DID INTEGER REFERENCES DOCTOR)
```

INSERT 5-7 RECORDS IN BOTH TABLE

```
SELECT * FROM DOCTOR  
SELECT * FROM PATIENT
```

1. Write a PL/SQL block to print senior citizen(maximum age of patient).
2. Write a PL/SQL block to print name of patient who is youngest.
3. WRITE A PL/SQL BLOCK TO COUNT TOTAL NUMBER OF RECORDS FROM PATIENT TABLE
4. WRITE A PL/SQL BLOCK TO DISPLAY MESSAGE OF PATIENT
E.G. TAKE INPUT AS PID, EXTRACT AGE AND DO THE FOLLOWING

TAKE AGE FROM TABLE AND BASED ON CONDITION DISPLAY THE RECORD

IF AGE >= 60 THEN DISPLAY- YOU ARE SENIOR CITIZEN

IF AGE >40 AND AGE <60 DISPLAY- YOU ARE MATURE CITIZEN

IF AGE >20 AND AGE <=40 DISPLAY- YOU ARE YOUTH CITIZEN

IF AGE >14 AND AGE <= 20 DISPLAY - YOU ARE TEENAGER

IF AGE <=14 DISPLAY YOU ARE CHILD

- 5 WRITE A PL/SQL BLOCK TO PRINT DEPT NAME OF DEPT ALONG WITH
EMPLOYEE NAME WHOSE DEPTNO = 2



ASSIGNMENT-5

Objective:

- To describe Explicit cursor- ALL records from table or selected records from table and manipulate data based on logic.

- 1 Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)

DEPT(deptid, dname, max_capacity)

Write a PL/SQL block to display all the records.
(use explicit cursor and cursor-For loop)

- 2 Write a PL/SQL block to display name, phone-no and designation of female employees.
- 3 Write a PL/SQL block to display highest paid first 5 employees.
- 4 ADD bonus field in EMP table.

Write a PL/SQL block to calculate bonus based on condition. After calculation of bonus of all the employees, display on screen as well as update it.

IF salary greater than 50k then bonus is 10% of salary

IF salary greater than 25k and less than 50k bonus is 15% of salary

If salary less than 25k then bonus is 20% of salary.

- 5 Assume that you have EMP and DEPT table with proper records:

Find out the name of employee who has experienced in the Electrical department more than 5 years.

- 6 Assume that you have EMP and DEPT table with proper records:

Write a PL-SQL block that will print the following report:

Id	NAme	Gender	Phone-no	Designation	Deptid	Deptname
1	Mahendra	M	9999222234	Manager	101	MCA
2	Virat	M	9873627789	Sales Manager	102	Civil
3	Rohit	M	8762349087	Clerk	101	MCA
4	Renu	F	7896734567	Accountant	103	Electrical

Total Records Printed : 4



ASSIGNMENT-6

Objective:

- To discuss Stored Procedure in PL/SQL from block or direct.

Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

- 1 Write a Procedure which display average salary of employee.
- 2 Write a Procedure which pass two parameters, calculate sum of two values and display it in procedure.
- 3 Write a Procedure which will display total no. of employees whose salary greater than 10000 and execute it.
- 4 Write a Procedure which will display mobile number of employee whose id is given by user. (pass as parameter)
- 5 Create a stored procedure that takes and input parameter as department- id and generate an output from the procedure as number of employees who are working in that department.
- 6 Write a procedure for following task:
When new employee joins in the institute, check if empid is present in the institute or not and the no. of already registered employees *not* exceeding the max capacity for that institute.
- 7 Write a procedure to display top five highest paid employees whose designation is Officer.
- 8 Write a procedure to display name of employee from emp table in a given format: (first name, middle name, last name), use CASE statement to have 4 options.
For example,
 - Display name in the format “ RLP “
 - Display name in the format “ R.L.Patel “
 - Display name in the format “ Rambhai L. Patel “
 - Display name in the format “ Rambhai Laxmanbhai Patel”
- 9 Enter the employee id from user. Check whether the name exists in table for corresponding id. If yes, Write a PL/SQL procedure to find total no of vowels from employee name. If no, display appropriate message.
- 10 To vote for Election, write a Procedure which will display the following details:
Eligible to vote(>18 years) || Not Eligible (less than 18 years)
Id & name of employee || id & Name of Employee



ASSIGNMENT-7

Objective:

- To discuss Stored Function and local function in PL/SQL from block.

- 1 Write a function to take two numbers from user and print summation of it.
- 2 Write a function whether given number is odd or even.
- 3 Write a function that returns total number of incomplete jobs, using table
JOB(jobid, type_of_job, status)
- 4 Write a function to which a birthdate is passed and it returns the age in terms of years as of today and a procedure/block which calls this function to display.
- 5 Write a function which displays the number of items whose weight fall between a given range for a particular color using table
ITEM(itemno, name, color, weight)
- 6 Create a function to return count of all employees who works in MCA department. This function is called in procedure which displays the output in proper format.

- 7 ISSUE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE)

After calculating the fine store the required information in the FINE table so that a report can later be printed out.

FINE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE, TOT_DAYS, FINE)

Write a procedure or function for the given a table, check which students have to pay fine and the amount of fine to be paid. A student can keep the book for 15 days. If the number of days has exceeded 15 then, the fine is calculated as follows:

Up to 7 days : 50 paise per day

8 – 15 days : Re 1 per day (to be counted from the 8th day onwards)

More than 15 : Re 1.5 per day (to be counted from the 16th day onwards)

- 8 NEWS_PAPER(news_id, name_of_paper, language, month_frequency, price)

Write a procedure that will display name of paper which is of lowest price.

Write a function which will count total number of newspaper exists language wise.

(English language : 999 papers

Gujarati language : 999 papers)

Hindi language : 999 papers)



ASSIGNMENT-8

Objective:

- To discuss Database Trigger in PL/SQL.

Assume above EMP and DEPT table exists with records.

- 1 Write a code which demonstrate simple trigger.
- 2 Create a trigger on EMP table that tracks the UPDATE on it. Each record being updated on it shall be tracked and kept as backup in EMP_TRAN table
- 3 Create a trigger on EMP table that tracks the DELETE on it. Every record being deleted from it shall be kept in EMP_TRAN_BACKUP table.
- 4 Create a trigger on EMP table that tracks UPDATE being performed on SAL column of the table. If the UPDATE is being performed between 2.30 pm to 3.30 pm then copy the effect on EMP_UPDATE_TRACK table
- 5 Write a trigger which restrict to enter or update the transaction on Saturday or Sunday
OR
Write a PL\SQL trigger which will not allow to insert records on saturday and sunday.
- 6 Write a trigger which restrict user to enter name in EMP table either starting with 'X' or is less than 2 letters
- 7 ADD Blood group in EMP table.

Write a trigger which only accept blood_gr in range(with +ve and -ve blood gr) of A ,B, AB, O.
- 8 Write a trigger in which accepts empid from user and check whether sal>65000 , don't allow to insert or update emp table.
- 9 Create a trigger which ensures that Emp name that to be inserted is in capital.
- 10 Write a trigger which deletes every record from result table if student table is empty.
student(studno,fnm,mnm,lnm,degree,bdth);
result(studno,math,eng ,ss,sci)
- 11 Write a trigger which accepts student records whose age is between 18 to 30.
- 12 Consider the table: (Eno, Ename, Basic, Hra, Da, Total)
Assume the following :
 - * Hra is given in Rupees
 - * DA is given as percentage as basic
 - * Total salary will be Basic + da*basic/100 + hra(i) Write a trigger so that the record on insertion contains basic salary in the range of 5000 to 18000;
DA in the range 16% to 32%; and HRA not more than 3000.



ASSIGNMENT-9

Objective:

- To discuss PL/SQL Parameterized cursor, Exceptional handling and package.

- 1 Item (Item code, item name, unit price)
Bill (bill number, bill date)
Bill_item (bill number, item code, number of units purchased)

Do the following:

- (a) For a given bill number, print the bill in the following format.

Bill no.:

Bill Date:

Sr. No.	Item code	Item name	Units purchased	unit price	total price
###	xxxxx	xxxxxxxxxxxxxxxxxxxxx	#####	####.##	#####.##

Bill Amount: #####.##

- (b) Add one more attribute named bill_amount in the Bill table. Update the Bill table to store bill amount.

- 2 **PATIENT(pid, pname, age, gender, doctor-id)**

Prepare following report for a Doctor information related to total patient age wise in his hospital.

Gender	Patient in Age Group					Total
	<u>1-12</u>	<u>13-25</u>	<u>25-50</u>	<u>50-100</u>	<u>Above 100</u>	
----	----	----	----	----	----	----
Total	-----	-----	-----	-----	-----	-----

Patient (pid, pame, age, gender , doctor-id)

Appoint(pcode, app_date, time , doctor_id)

Doctor(doctor_id, name, specialization)

Prepare a report from above table taking date, shift, , doctor_id as parameters.

Shift is Morning if time are from 9.00 Am to 1.00 pm and

Shift is Evening if time are from 5.00 to 9.30 pm.

APPOINTMENT SCHEDULE

Date :

Doctor name :

Shift :

Specialization :

Time	Patient name	Age
9.00		
9.10		
9.20		
.....		



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Total no. of patients : _____

- 3 ALBUM(album_no, album_name, album_type, release_date, singer_name, no_songs)

Write a procedure which takes parameter as singer name and print the following album details:

Singer name : _____

Album no	Album name	Type	Release Date	Number of Songs
99	Xx	Xx	Xx	99
99	Xx	Xx	Xx	99

- 4 Create table and insert proper data.

Tour_Details(tour_no, start_date, group_size, source, destination, vehicle_no, driver_name)

Write a PL/SQL block which will print tour details, a driver is going to take it. (pass driver_no as parameter)

Driver Name : _____ Vehicle number : _____

Tour Details

Source	Destination	Group size	Start date	Total days
Xxx	Xxx	999	Xxx	999

- 5 CUST(cno, accno, name, add, city , phone, email)
ACCOUNT(accno, amount)

Write a stored procedure for banking purpose which perform following Task :

1. Create a new account (insert)
2. Deposit funds (update)
3. Check account balance (view balance)
4. Close the account (delete)

- 6 Create table and insert proper data.

TRAIN_MAST(train_no, train_name, arrival_time, departure_time, source, destination)

Write a procedure which will print all train details going from Baroda to Bombay
Write a function which will print arrival time and departure time for a given train. (pass train no as a parameter)



ASSIGNMENT-10

1 Given hospital schema

Patient(pat_code, pname, birthdate, gender)

Dentist(dcode, dname, qualification, area, city, experience)

Treatment(dcode, pat_code, treatment_date, treat_details)

Write a procedure or Function to do the following queries: (USE Switch case)

1. Count no. of dentist from 'lalbaug' area in baroda city.
2. Display details of dentist with 'MDS' as qualification.
3. Display details of treatment given to patient no.'P101'.
4. Display total number of patient who came for treatment as on '12-Feb-2013'.
5. Display area wise list of dentist who has maximum experience.
6. Display details of youngest dentist.
7. Display dentist details along with their patient.
8. Display details of all female dentist.
9. Count total number of male dentist who live in 'ahmedabad'.
10. Display patient name without duplicate who come for treatment.
11. Display details of patient who has not given any treatment.

2 Consider the EMPLOYEE schema as :

EMPMAS(empno, name, pfno, empbasic, deptno, designation)

HOLIDAY(year, month, holidays, weekoff)

EMPTRAN(empno, month, year, presence, loan(amt))

BANK(accno, empno, bank name, bank branch)

Rules : HRA = 15% of basic

DA = 50% of basic

Medical = 100

PF = 8.33% of basic

Salary is given for (attendance + holidays + weekoff) days.

Print salary slip in foll. Format.

SALARY SLIP of Month : _____ Year _____			
Accno :	Name of EMP	Designation	Dept
Bank name :	_____	Branch : _____	Dt of Join _____
Presence :	_____	Absent days : _____	Sal days : _____
EARNINGS :		DEDUCTIONS	
BASIC :		LOAN :	
DA :		PF :	
HRA :		Prof tax : 100	
MEDICAL : 100			



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TOTAL earnings: _____

TOTAL Deductions: _____

NET PAY : _____

- 3 Write a code which demonstrate user defined exception.
- 4 Write a code which demonstrate package in PL/SQL.

=====THE END, NO the Journey doesn't end here !!!=====



1. Create circuit of AND gate using TINKERCAD.
2. Create circuit of OR gate using TINKERCAD.
3. Create circuit of XOR gate using TINKERCAD.
4. Create circuit of Half Adder using TINKERCAD.
5. Create circuit of Half Adder using TINKERCAD.
6. Write an 8051 C program to send values 00 to FF to port 1.
7. Write an 8051 C program to send hex value for ASCII character 0,1,2,3,4,5,A,B,C,D to port 1.
8. Write an 8051 C program to toggle all bits of P1 continuously.
9. Write an 8051 C program to toggle bit D0 of port 1 50000 times.
10. Write an 8051 C program to send value -4 to +4 to port 2.
11. Write an 8051 C program to toggle only bit P2.4 continuously without disturbing other bits,
12. Write an 8051 C program to get a byte of data from P0 if it is less than 100 send it to P1 otherwise send it to P2.
13. Write an 8051 C program to monitor bit P1.5. If it is high send 55H to P0 else send AAH to P2.
14. A door sensor is connected to P1.1 and buzzer is connected to P1.7. Write an 8051 C program to monitor door sensor, and when it opens sound the buzzer. Sound the buzzer by sending square wave.
15. The data pins of LCD are connected to P1. Write an 8051 C program to send "The earth is but one country" to this LCD.
16. Write an 8051 C program to toggle all bits of P0, P1 and P2 continuously. Use the sfr keyword to declare port address.
17. Write an 8051 C program to turn bit P1.5 on and off 50000 times. Use sbit keyword.
18. Write an 8051 C program to get status of bit P1.0, save it and send it to P2.7 continuously. Use sbit keyword.
19. Write an 8051 C program to toggle all bits of P0 and P2 using inverting operator.
20. Write an 8051 C program to toggle all bits of P0 and P2 using Ex-OR operator.
21. Write an 8051 C program to read the P1.0 and P1.1 and issue an ASCII character to P0 according to following table.

P1.1	P1.0	Function
0	0	Send '0' to P0
0	1	Send '1' to P0
1	0	Send '2' to P0
1	1	Send '3' to P0



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22. Write an 8051 C program to convert packed BCD 0x29 to ASCII and display bytes on P1 and P2.
23. Write an 8051 C program to convert ASCII digits '4' and '7' to packed BCD and display them on P1
24. Write an 8051 C program to calculate checksum byte for following data
25H, 62H, 3FH, 52H
25. Write an 8051 C program to check whether data is good or bad. If data is good, send 'G' to P0 otherwise send 'B' to P0. (assume data from program 4)
26. Write an 8051 C program to convert 11111101 (FD hex) to decimal and display digits on P0, P1 and P2.
27. Write an 8051 C program to toggle only bit 1.5 continuously every 50 ms. Use timer 0, mode 1 to create the delay.
28. Write an 8051 C program to toggle all bits of P2 continuously every 500 ms. Use timer 1, mode 1 to create the delay.
29. Write an 8051 C program to create frequency of 2500 hz on pin P2.7. Use timer 1, mode 2 to create the delay.
30. Write an 8051 C program to generate a square wave with an ON time of 3 ms and OFF time of 10 ms on all pins of Port 0. Assume crystal frequency of 22 MHz.